

The Holomorphic Rotational KMS Extension Theorem: Index Four, the Canonical Expectation, and the TFPT Finite Shadow

Proof memorandum

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Abstract

Let $\mathcal{A} \subset \mathcal{B}$ be an irreducible finite-index inclusion of conformal nets on S^1 , let $\mathcal{A} = \mathcal{B}^{\mathbb{Z}_4}$ for a proper gauge action commuting with rotations, and suppose that $\mathcal{B}$ is holomorphic, with $\mu_{\mathcal{B}} = 1$ and central charge 8. We prove the strongest correct form of the proposed KMS extension statement. At angular inverse temperature $\beta > 0$, the vacuum-sector Gibbs state of $\mathcal{B}$ restricts to a rotational KMS state ω_{β} of $\mathcal{A}$. The group average

$$E = \frac{1}{4} \sum_{r=0}^3 \alpha_r$$

is the unique normal faithful conditional expectation $\mathcal{B}(I) \rightarrow \mathcal{A}(I)$ on every interval, and the extension is $\tilde{\omega}_{\beta} = \omega_{\beta} \circ E$. It is the unique $\mathbb{Z}_4$ -invariant state extension, even before the KMS condition is imposed. Finally, the Kawahigashi–Longo–Müger sector theorem together with the rotational KMS classification of Longo–Tanimoto implies that a holomorphic conformal net has only one rotational β -KMS state. Hence every rotational KMS extension is automatically $\mathbb{Z}_4$ -invariant and equals $\omega_{\beta} \circ E$.

Two qualifications are essential. The vacuum *vector* state is a rotation ground state, not a finite-temperature rotational KMS state; “vacuum KMS” below means the vacuum-*sector* Gibbs state. Also, $\mu = 1$ eliminates DHR sectors but does not by itself classify thermal states: the Longo–Tanimoto theorem is the necessary analytic bridge. We quote in full the precise forms of Takesaki’s expectation theorem, the finite-index/Q-system theorem, the KLM μ -index theorem, Araki’s CAR KMS theorem, and the rotational KMS classification, and verify their hypotheses against the frozen TFPT finite-collar data. The final section isolates the remaining continuum identification in one box.

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1 Setting and correction of terminology

1.1 Conformal-net hypotheses

Let $I \mapsto \mathcal{B}(I)$ be an irreducible local conformal net on S^1 in its vacuum representation $(\mathcal{H}_{\mathcal{B}}, U_{\mathcal{B}}, \Omega_{\mathcal{B}})$. Let $G = \mathbb{Z}_4$ act properly by vacuum-preserving internal symmetries α_r , meaning

$$\alpha_r(\mathcal{B}(I)) = \mathcal{B}(I), \quad \alpha_r \operatorname{Ad} U_{\mathcal{B}}(g) = \operatorname{Ad} U_{\mathcal{B}}(g) \alpha_r$$

for every interval I , every conformal transformation g , and $r \in \mathbb{Z}_4$. Put

$$\mathcal{A}(I) = \mathcal{B}(I)^G.$$

We assume that the local inclusion is irreducible and has Jones index four:

$$[\mathcal{B}(I) : \mathcal{A}(I)] = 4, \quad \mathcal{A}(I)' \cap \mathcal{B}(I) = \mathbb{C}\mathbf{1}. \quad (1.1)$$

For a proper outer action of a finite group on a factor, the first identity follows from the fixed-point theorem; it is also included explicitly among our hypotheses so no outerness convention is hidden.

The net $\mathcal{B}$ is assumed holomorphic in the operator-algebraic sense: it is completely rational and

$$\mu_{\mathcal{B}} = 1. \quad (1.2)$$

The central charge $c = 8$ identifies the intended endpoint as the $(E_8)_1$ net, but is not needed for KMS uniqueness once (1.2) is known.

1.2 The rotational KMS state

Let $L_0^{\mathcal{B}} \geq 0$ be the conformal Hamiltonian and let $\tau_t = \operatorname{Ad} e^{itL_0^{\mathcal{B}}}$ be the rotation dynamics on the locally normal universal algebra $C_{\text{in}}^*(\mathcal{B})$. For $\beta > 0$ put

$$\Phi_{\beta}^{\mathcal{B}}(x) = \frac{\operatorname{Tr}_{\mathcal{H}_{\mathcal{B}}}(e^{-\beta L_0^{\mathcal{B}}} x)}{\operatorname{Tr}_{\mathcal{H}_{\mathcal{B}}}(e^{-\beta L_0^{\mathcal{B}}})}. \quad (1.3)$$

For a completely rational conformal net the character is finite, and the rotational KMS classification quoted below in fact recovers the trace-class property from any extremal KMS state. In the concrete $(E_8)_1$ case it is also immediate from the character E_4/η^8 .

Definition 1.1 (Geometric rotational state used here). *The state on the subnet is*

$$\omega_\beta := \Phi_\beta^{\mathcal{B}}|_{C_{\text{in}}^*(\mathcal{A})}. \quad (1.4)$$

At the canonical angular normalization we set $\beta = \beta_{\text{angle}} = 2\pi$.

Remark 1.2 (Ground state versus rotational KMS state). The vector functional $\langle \Omega_{\mathcal{B}}, \cdot \Omega_{\mathcal{B}} \rangle$ is a ground state for rotations. It is not the finite-temperature state (1.3). The geometric KMS state on the real line for translations/dilations and the rotational Gibbs state on the circle are likewise different constructions. The theorem below concerns the latter, because its unconditional classification by superselection sectors is available in exactly this setting.

2 Classical theorems used, stated in full

2.1 Araki's CAR KMS theorem

Theorem 2.1 (Araki: unique quasi-free CAR equilibrium state). *Let $\mathfrak{h}$ be a complex Hilbert space, let $h = h^*$ be a possibly unbounded one-particle Hamiltonian, and let τ_t^h be the strongly continuous Bogoliubov dynamics on $\text{CAR}(\mathfrak{h})$ determined on generators by*

$$\tau_t^h(a(f)) = a(e^{ith}f).$$

For every $\beta > 0$ there is a gauge-invariant quasi-free state $\varphi_{\beta,h}$ whose two-point function is

$$\varphi_{\beta,h}(a^*(f)a(g)) = \langle g, C_{\beta,h}f \rangle, \quad C_{\beta,h} = (\mathbf{1} + e^{\beta h})^{-1}, \quad (2.1)$$

and whose higher even correlation functions are the determinants prescribed by the CAR Wick rule, while odd correlation functions vanish. This state is a β -KMS state for τ^h . In the factorial self-dual CAR representation (in particular when the covariance has no 0 or 1 eigenvalue in the finite-dimensional case), it is the unique β -KMS state for that Bogoliubov dynamics. Thus the KMS state is completely determined by its two-point covariance.

This is the KMS uniqueness theorem in H. Araki, *On quasifree states of CAR and Bogoliubov automorphisms*, Publ. Res. Inst. Math. Sci. **6** (1970/71), 385–442, Theorem 3 and Eq. (6.9), translated from the self-dual notation to the ordinary gauge-invariant CAR notation. Araki's factor proviso is recorded rather than suppressed.

2.2 Takesaki's conditional-expectation theorem

Theorem 2.2 (Takesaki: modular invariance criterion). *Let M be a von Neumann algebra, let ϕ be a faithful normal semifinite weight on M , and let $N \subset M$ be a von Neumann subalgebra such that $\phi|_N$ is semifinite. Write $\sigma^\phi = (\sigma_t^\phi)_{t \in \mathbb{R}}$ for the modular automorphism group of ϕ . The following are equivalent:*

- (i) *N is globally invariant under the modular group, $\sigma_t^\phi(N) = N$ for every $t \in \mathbb{R}$;*
- (ii) *there exists a normal faithful conditional expectation $E_\phi : M \rightarrow N$ such that $\phi \circ E_\phi = \phi$ on M_+ .*

When these conditions hold, E_ϕ is unique among the normal conditional expectations preserving ϕ , it commutes with the modular group,

$$E_\phi \sigma_t^\phi = \sigma_t^{\phi|N} E_\phi, \quad (2.2)$$

and the modular group of the restricted weight is the restriction:

$$\sigma_t^{\phi|N} = \sigma_t^\phi|_N. \quad (2.3)$$

This is M. Takesaki, *Conditional expectations in von Neumann algebras*, J. Funct. Anal. **9** (1972), 306–321, and *Theory of Operator Algebras II*, Springer, 2003, Theorem IX.4.2.

2.3 Finite index, Radon–Nikodym uniqueness, and Q-systems

Theorem 2.3 (Finite-index expectation and Q-system package). *Let $N \subset M$ be an inclusion of infinite factors with finite index and separable predual.*

- (a) *There is a normal faithful conditional expectation $E_0 : M \rightarrow N$ of finite index. Fix one such expectation. Every other normal faithful conditional expectation $F : M \rightarrow N$ is of the form*

$$F(x) = E_0(h^{1/2} x h^{1/2}), \quad x \in M, \quad (2.4)$$

for a positive invertible $h \in N' \cap M$ satisfying $E_0(h) = \mathbf{1}$. Conversely every such h defines an expectation. In particular, if $N' \cap M = \mathbb{C}\mathbf{1}$, then the normal faithful conditional expectation is unique.

- (b) *Let $\iota : N \hookrightarrow M$ be the inclusion morphism and $\bar{\iota} : M \rightarrow N$ a conjugate. The dual canonical endomorphism $\theta = \bar{\iota}\iota \in \text{End}(N)$, together with standard conjugacy intertwiners*

$$w \in \text{Hom}(\mathbf{1}_N, \theta), \quad x \in \text{Hom}(\theta, \theta^2),$$

is a standard Q-system: w is a unit, x is associative and unital with w , and x obeys the Frobenius and standardness relations. Conversely, every standard Q-system in $\text{End}(N)$ reconstructs, up to isomorphism fixing N , a finite-index extension $N \subset M$. Its dimension is

$$d(\theta) = [M : N]. \quad (2.5)$$

If the endomorphisms lie in the braided DHR category of a conformal net, the reconstructed extension is local exactly when the Q-system is commutative with respect to the braiding.

- (c) *If M carries an outer action of a finite group G and $N = M^G$, then*

$$E_0(x) = \frac{1}{|G|} \sum_{g \in G} \alpha_g(x), \quad [M : N] = |G|. \quad (2.6)$$

The dual canonical endomorphism is the regular Q-system. For $G = \mathbb{Z}_4$ it is the sum of the four one-dimensional character sectors and has dimension four.

Part (a) is the Radon–Nikodym description of expectations in the finite-index theory of H. Kosaki and R. Longo; see R. Longo, *Index of subfactors and statistics of quantum fields I*, Commun. Math. Phys. **126** (1989), 217–247, especially the uniqueness/minimal-index discussion, and F. Hiai, *Minimum index for subfactors and entropy II*, J. Math. Soc. Japan **43** (1991), 347–379. Parts (b) and (c) are the subfactor/Q-system and fixed-point forms used in R. Longo and K.-H. Rehren, *Nets of subfactors*, Rev. Math. Phys. **7** (1995), 567–597, together with the local-extension formulation of J. Böckenhauer and D. E. Evans, *Modular invariants, graphs and α -induction for nets of subfactors I*, Commun. Math. Phys. **197** (1998), 361–386.

2.4 The KLM μ -index theorem

Theorem 2.4 (Kawahigashi–Longo–Müger). *Let $\mathcal{N}$ be a local irreducible conformal net satisfying the split property and strong additivity, and suppose that its two-interval inclusion*

$$\mathcal{N}(E) \subset \widehat{\mathcal{N}}(E) := \mathcal{N}(E')'$$

has finite index. Then $\mathcal{N}$ is completely rational; it has finitely many inequivalent irreducible DHR sectors $\rho_0 = \mathbf{1}, \rho_1, \dots, \rho_n$, all of finite statistical dimension, every locally normal representation is a direct sum of these sectors, the braiding is non-degenerate, and

$$\mu_{\mathcal{N}} = [\widehat{\mathcal{N}}(E) : \mathcal{N}(E)] = \sum_{j=0}^n d(\rho_j)^2. \quad (2.7)$$

The two-interval inclusion is the Longo–Rehren inclusion associated with the full sector category. If $\mathcal{N} \subset \mathcal{M}$ is a finite-index irreducible local extension of completely rational nets, then

$$\mu_{\mathcal{M}} = \frac{\mu_{\mathcal{N}}}{[\mathcal{M} : \mathcal{N}]^2}. \quad (2.8)$$

Consequently, $\mu_{\mathcal{M}} = 1$ if and only if the vacuum is the only irreducible DHR sector of $\mathcal{M}$.

This is Y. Kawahigashi, R. Longo and M. Müger, *Multi-interval subfactors and modularity of representations in conformal field theory*, Commun. Math. Phys. **219** (2001), 631–669: the two-interval/Longo–Rehren identification, the equality of μ with the global sector dimension, and the extension-index formula are the results used here. The final equivalence follows from (2.7), since $d(\rho_0) = 1$ and every nonzero statistical dimension is at least one.

2.5 Rotational KMS classification

Theorem 2.5 (Longo–Tanimoto). *Let $\mathcal{N}$ be a completely rational conformal net and let $C_{\text{ln}}^*(\mathcal{N})$ be its locally normal universal C^* -algebra. Then*

$$C_{\text{ln}}^*(\mathcal{N}) \cong \bigoplus_{j=0}^n B(\mathcal{H}_j), \quad (2.9)$$

where the summands correspond bijectively to the inequivalent irreducible DHR representations ρ_j . Every extremal rotational β -KMS state is the Gibbs state in one irreducible representation:

$$\Phi_{j,\beta}(x) = \frac{\text{Tr}_{\mathcal{H}_j}(e^{-\beta L_0^{(j)}} \rho_j(x))}{\text{Tr}_{\mathcal{H}_j}(e^{-\beta L_0^{(j)}})}. \quad (2.10)$$

In particular $e^{-\beta L_0^{(j)}}$ is trace class for every sector that supports such a state. Every rotational β -KMS state is a convex combination of the extremal sector Gibbs states. Hence, if the vacuum is the only irreducible sector, the vacuum-sector Gibbs state is the unique rotational β -KMS state.

Equation (2.9) is the completely rational universal algebra theorem of Carpi–Conti–Hillier–Weiner used in R. Longo and Y. Tanimoto, *Rotational KMS states and type I conformal nets*, Commun. Math. Phys. **357** (2018), 249–266, Theorems 3.1, 3.2 and 3.7. The last sentence is an immediate specialization to a one-summand algebra.

3 Verification of the hypotheses

3.1 The index-four fixed-point Q-system

For each interval set $N = \mathcal{A}(I)$ and $M = \mathcal{B}(I)$. Conformal-net local algebras are factors; (1.1) gives finite index and trivial relative commutant. Theorem 2.3 therefore applies. The group average is

$$E_I(x) = \frac{1}{4}(x + \alpha_1(x) + \alpha_2(x) + \alpha_3(x)). \quad (3.1)$$

It is normal, unital, positive, faithful, idempotent, and $\mathcal{A}(I)$ -bimodular. For example, if $a, b \in \mathcal{A}(I)$, then

$$E_I(axb) = \frac{1}{4} \sum_{r=0}^3 a\alpha_r(x)b = aE_I(x)b. \quad (3.2)$$

Faithfulness follows because $x \geq 0$ and $E_I(x) = 0$ imply that the sum of four positive operators is zero, hence $x = 0$. Compatibility under interval inclusion is immediate from the same formula, so the E_I define a net expectation and a global group average

$$E : C_{\text{ln}}^*(\mathcal{B}) \longrightarrow C_{\text{ln}}^*(\mathcal{A}). \quad (3.3)$$

The regular $\mathbb{Z}_4$ Q-system has

$$\theta \simeq \rho_0 \oplus \rho_1 \oplus \rho_2 \oplus \rho_3, \quad d(\rho_r) = 1, \quad d(\theta) = 4. \quad (3.4)$$

The frozen TFPT simple-current arithmetic $h(J^r) = 1$ for $r = 1, 2, 3$ verifies the integer-spin/locality condition for the intended index-four extension. The arithmetic cited in v973 is

$$\mu_{\mathcal{A}} = 16, \quad [\mathcal{B} : \mathcal{A}] = 4, \quad \mu_{\mathcal{B}} = \frac{16}{4^2} = 1. \quad (3.5)$$

This is precisely the KLM extension formula (2.8); the Python computation is a finite exact check of its numerical inputs, not a replacement for the theorem.

3.2 The frozen CAR covariance

The finite collar used by v995 has eight complex fermion modes, hence

$$\text{CAR}_8 \simeq M_{2^8}(\mathbb{C}) = M_{256}(\mathbb{C}).$$

In dimensionless modular units the one-particle Hamiltonian is

$$h_{\text{collar}} = \Delta \text{diag}(1, 2, \dots, 8), \quad \Delta = 6 \log(3/2), \quad (3.6)$$

and the script sets $\beta = 1$. Therefore

$$q = e^{-\Delta} = (2/3)^6 = \frac{64}{729}, \quad C_j = \frac{q^j}{1 + q^j} = \frac{1}{1 + e^{j\Delta}}, \quad j = 1, \dots, 8. \quad (3.7)$$

This is exactly Araki's covariance (2.1). All eigenvalues lie strictly between 0 and 1, so the finite CAR representation is factorial and the density matrix is faithful. On the occupation basis $n = (n_1, \dots, n_8) \in \{0, 1\}^8$,

$$p(n) = Z^{-1} q^{\sum_{j=1}^8 j n_j}, \quad Z = \prod_{j=1}^8 (1 + q^j). \quad (3.8)$$

Thus all 256 weights are positive, and

$$\frac{p(n + e_j)}{p(n)} = q^j$$

whenever $n_j = 0$. These are the exact detailed-balance relations checked in v995. The angular convention $\beta_{\text{angle}} = 2\pi$ is recovered by absorbing 2π into the modular Hamiltonian: $\beta h = (2\pi)H_{\text{angle}}$.

Remark 3.1 (What Araki proves here). Araki's theorem proves uniqueness of the KMS state of the full CAR dynamics and its determination by (3.7). Restricting that state to the even/bilinear $\text{SO}(16)_1$ subnet selects the vacuum-sector thermal state. It does *not* say that the observable $\text{SO}(16)_1$ net has only one rotational KMS state: that net has four DHR sectors and hence four extremal sector Gibbs states. Holomorphic extension to $\mathcal{B}$ is what removes this sector choice.

3.3 Holomorphy and thermal classification

By (1.2) and Theorem 2.4,

$$1 = \mu_{\mathcal{B}} = \sum_{\rho \in \text{Irr}(\mathcal{B})} d(\rho)^2. \quad (3.9)$$

The vacuum already contributes one, so there can be no other irreducible DHR sector. Theorem 2.5 then has one Gibbs state and no other extreme point. This verifies the analytic hypothesis needed for unconditional rotational KMS uniqueness.

4 The holomorphic KMS extension theorem

Theorem 4.1 (Holomorphic rotational KMS extension). *Let $\mathcal{A} \subset \mathcal{B}$ satisfy the hypotheses of Section 1: for every interval I , $\mathcal{A}(I) \subset \mathcal{B}(I)$ is an irreducible index-four inclusion, $\mathcal{A} = \mathcal{B}^{\mathbb{Z}_4}$ for a proper internal gauge action commuting with rotations, and $\mathcal{B}$ is a holomorphic conformal net with $\mu_{\mathcal{B}} = 1$ and $c = 8$. Let $\beta > 0$ and let ω_{β} be the restriction (1.4) of the vacuum-sector rotational Gibbs state. Then:*

(a) ω_{β} has a rotational β -KMS extension to $\mathcal{B}$.

(b) The maps

$$E_I = \frac{1}{4} \sum_{r=0}^3 \alpha_r : \mathcal{B}(I) \rightarrow \mathcal{A}(I)$$

are the unique normal faithful conditional expectations. They form a net expectation E , and

$$\tilde{\omega}_{\beta} = \omega_{\beta} \circ E = \Phi_{\beta}^{\mathcal{B}}. \quad (4.1)$$

This is the unique $\mathbb{Z}_4$ -invariant state extension of ω_{β} , hence in particular the unique $\mathbb{Z}_4$ -invariant rotational KMS extension.

(c) Every rotational β -KMS extension of ω_{β} is $\mathbb{Z}_4$ -invariant. Consequently $\tilde{\omega}_{\beta}$ is the unique rotational KMS extension without an invariance qualifier.

Proof. Part (a). The Gibbs functional (1.3) is a rotational β -KMS state on $C_{\text{ln}}^*(\mathcal{B})$ by Theorem 2.5, or directly by the cyclicity of the trace on analytic elements:

$$\begin{aligned} \Phi_{\beta}^{\mathcal{B}}(x\tau_{t+i\beta}(y)) &= Z_{\beta}^{-1} \text{Tr}(e^{-\beta L_0} x e^{itL_0} e^{-\beta L_0} y e^{\beta L_0} e^{-itL_0}) \\ &= Z_{\beta}^{-1} \text{Tr}(e^{-\beta L_0} \tau_t(y)x) = \Phi_{\beta}^{\mathcal{B}}(\tau_t(y)x). \end{aligned} \quad (4.2)$$

Its restriction to a τ -invariant C^* -subalgebra remains KMS: the same strip-analytic functions work for $x, y \in C_{\text{ln}}^*(\mathcal{A})$. By definition that restriction is ω_β . Hence $\Phi_\beta^{\mathcal{B}}$ is the required extension.

Part (b): expectation and its uniqueness. Section 3.1 proves directly that the group average E_I is a normal faithful conditional expectation. If $F_I : \mathcal{B}(I) \rightarrow \mathcal{A}(I)$ is any other normal faithful conditional expectation, the Radon–Nikodym part of Theorem 2.3 gives

$$F_I(x) = E_I(h^{1/2}xh^{1/2})$$

with $h \in \mathcal{A}(I)' \cap \mathcal{B}(I)$. Irreducibility gives $h = \lambda \mathbf{1}$. Since $F_I(\mathbf{1}) = \mathbf{1} = E_I(\mathbf{1})$, $\lambda = 1$, and $F_I = E_I$. This also shows why the bimodule property in the definition of a conditional expectation is load-bearing: without it, the Radon–Nikodym theorem does not apply, exactly as the finite mutants in v993 demonstrate.

The Gibbs state is gauge invariant. Indeed the internal symmetry commutes with L_0 and is unitarily implemented, so

$$\begin{aligned} \Phi_\beta^{\mathcal{B}}(\alpha_r(x)) &= Z_\beta^{-1} \text{Tr}(e^{-\beta L_0} V_r x V_r^*) \\ &= Z_\beta^{-1} \text{Tr}(V_r^* e^{-\beta L_0} V_r x) = \Phi_\beta^{\mathcal{B}}(x). \end{aligned} \quad (4.3)$$

Consequently

$$\Phi_\beta^{\mathcal{B}}(E(x)) = \frac{1}{4} \sum_{r=0}^3 \Phi_\beta^{\mathcal{B}}(\alpha_r(x)) = \Phi_\beta^{\mathcal{B}}(x). \quad (4.4)$$

Because $E(x) \in \mathcal{A}$ and $\Phi_\beta^{\mathcal{B}}|_{\mathcal{A}} = \omega_\beta$,

$$\Phi_\beta^{\mathcal{B}}(x) = \Phi_\beta^{\mathcal{B}}(E(x)) = \omega_\beta(E(x)),$$

which proves (4.1).

Now let ψ be any $\mathbb{Z}_4$ -invariant state on $\mathcal{B}$ whose restriction to $\mathcal{A}$ is ω_β . No KMS hypothesis is needed:

$$\psi(x) = \frac{1}{4} \sum_{r=0}^3 \psi(\alpha_r(x)) = \psi(E(x)) = \omega_\beta(E(x)). \quad (4.5)$$

Thus $\psi = \omega_\beta \circ E$.

The modular-theory chain. We spell out the Takesaki step because it is frequently reversed incorrectly. In the GNS representation of the faithful KMS state, the modular group and the physical dynamics are related, with the present KMS convention, by

$$\sigma_s^{\Phi_\beta^{\mathcal{B}}}(\pi_\Phi(x)) = \pi_\Phi(\tau_{-\beta s}(x)). \quad (4.6)$$

Rotations preserve the subnet, hence

$$\sigma_s^{\Phi_\beta^{\mathcal{B}}}(\pi_\Phi(\mathcal{A}(I))) = \pi_\Phi(\mathcal{A}(I)).$$

Takesaki's theorem now gives the unique $\Phi_\beta^{\mathcal{B}}$ -preserving normal faithful expectation. By (4.4), the group average is such an expectation, so it is the Takesaki expectation. Moreover,

$$\sigma_s^{\omega_\beta} = \sigma_s^{\Phi_\beta^{\mathcal{B}}}|_{\mathcal{A}(I)} = \tau_{-\beta s}|_{\mathcal{A}(I)}, \quad E_I \sigma_s^{\Phi_\beta^{\mathcal{B}}} = \sigma_s^{\omega_\beta} E_I. \quad (4.7)$$

Notice the logical order: existence of the Gibbs extension gives a state on $\mathcal{B}$; modular invariance then identifies its state-preserving expectation. Takesaki's criterion alone would not manufacture a KMS extension from an arbitrary state on $\mathcal{A}$.

Part (c): removal of the invariance qualifier. Let ψ be a rotational β -KMS state on $C_{\text{ln}}^*(\mathcal{B})$ extending ω_β . By Theorem 2.4 and (3.9), $\mathcal{B}$ has only the vacuum irreducible DHR sector.

Theorem 2.5 says that every rotational KMS state is a convex combination of Gibbs states in irreducible sectors. There is only one such state, $\Phi_\beta^\mathcal{B}$, so

$$\psi = \Phi_\beta^\mathcal{B}.$$

Equation (4.3) makes this state $\mathbb{Z}_4$ -invariant, and part (b) then gives $\psi = \omega_\beta \circ E$. This proves unconditional uniqueness. $\square$

5 Where the mixing directions went

5.1 Relative-commutant mixing

Fix an interval and write $N = \mathcal{A}(I)$, $M = \mathcal{B}(I)$. An internal multiplicity mixer for the inclusion is an operator $T \in N' \cap M$. By irreducibility,

$$N' \cap M = \mathbb{C}\mathbf{1}, \tag{5.1}$$

so $T = \lambda\mathbf{1}$. A normalized perturbation cannot use this scalar to create a new state direction. Explicitly, if $T = T^*$ and

$$\psi_\varepsilon(x) = \frac{\Phi_\beta^\mathcal{B}((\mathbf{1} + \varepsilon T)x(\mathbf{1} + \varepsilon T))}{\Phi_\beta^\mathcal{B}((\mathbf{1} + \varepsilon T)^2)},$$

then (5.1) makes numerator and denominator carry the same scalar factor, and

$$\psi_\varepsilon = \Phi_\beta^\mathcal{B}. \tag{5.2}$$

Thus every mixing direction that would be implemented by an operator commuting with the subnet is trivial.

5.2 Superselection-sector mixing

A different possible mixing is a convex combination of thermal states in inequivalent irreducible representations. KLM gives the sector count through

$$\mu_\mathcal{B} = \sum_j d(\rho_j)^2.$$

Holomorphy makes the sum equal one, so only $\rho_0 = \mathbf{1}$ remains. Longo–Tanimoto identifies the extreme rotational KMS states with the sector Gibbs states. Therefore there is no second thermal sector into which a state can mix.

These are two distinct exclusions:

$\mathcal{A}(I)' \cap \mathcal{B}(I) = \mathbb{C}\mathbf{1}$ removes local multiplicity mixing.
 $\mu_\mathcal{B} = 1$, together with the Longo–Tanimoto classification, removes thermal-sector mixing.

Neither exclusion should be substituted for the other. In particular, holomorphy alone is a statement about DHR sectors, not an abstract theorem about all possible dynamics.

6 TFPT corollary and the P1 reading

6.1 The finite matrix shadow

Corollary 6.1 (Interpretation of v995). *Assume that the TFPT collar scaling limit realizes the net inclusion of Theorem 4.1 with the stated rotation dynamics. Then the exact finite results of v995 are finite-dimensional shadows of the two uniqueness mechanisms:*

- (i) *on $\text{CAR}_8 = M_{256}$, the KMS equations have rank $256^2 - 1 = 65535$ on the state tangent and kernel zero;*
- (ii) *all 33 selected same- $\mathbb{Z}_4$ -charge, unequal-modular-energy Hermitian directions fail stationarity and are killed by KMS;*
- (iii) *on the finite crossed product, fixing the grade-zero restriction leaves*

$$3 \cdot 256^2 = 196608$$

real linear extension coordinates, and invariance under $E = (1/4) \sum_r \alpha_r$ kills all of them;

- (iv) *covariance is redundant after the full finite-factor KMS condition has been imposed.*

Proof. Items (i)–(iv) are exactly the finite linear-algebra statements of v995. The continuum interpretation is conditional on the scaling-limit identification because finite matrix rank does not by itself prove factoriality or classify locally normal KMS states of a type-III net. Under that identification, item (iii) is the finite Fourier version of (4.5), while items (i), (ii), and (iv) are the finite CAR instance of Theorem 2.1. $\square$

6.2 Equidistribution over four deck channels

The exact statement needed for the P1 reading is elementary but has one hypothesis that must be visible.

Lemma 6.2 (Transitive four-channel equidistribution). *Let p_0, p_1, p_2, p_3 be orthogonal projections in a response algebra with $\sum_r p_r = \mathbf{1}$, and suppose a $\mathbb{Z}_4$ action is transitive:*

$$\alpha_s(p_r) = p_{r+s \bmod 4}. \quad (6.1)$$

If ψ is $\mathbb{Z}_4$ -invariant, then

$$\psi(p_0) = \psi(p_1) = \psi(p_2) = \psi(p_3) = \frac{1}{4}. \quad (6.2)$$

If a normalized total modular response is $R_{\text{tot}} = 1/\beta_{\text{angle}}$ and is additive over these channels, then each channel carries

$$R_r = \frac{1}{4\beta_{\text{angle}}}. \quad (6.3)$$

Proof. For any r, s , invariance and transitivity give

$$\psi(p_{r+s}) = \psi(\alpha_s(p_r)) = \psi(p_r).$$

All four values are therefore a common number w . Normalization gives $1 = \psi(\mathbf{1}) = \sum_r \psi(p_r) = 4w$, hence $w = 1/4$. Additivity of the response then gives $R_r = R_{\text{tot}}/4$. $\square$

Corollary 6.3 (The theorem-conditional P1 identity). *Under Theorem 4.1 and the response-resolution hypothesis of Lemma 6.2, the canonical geometric normalization*

$$\beta_{\text{angle}} = 2\pi \quad (6.4)$$

gives

$$c_3 = R_r = \frac{1}{4\beta_{\text{angle}}} = \frac{1}{4 \cdot 2\pi} = \frac{1}{8\pi}, \quad 4\beta_{\text{angle}}c_3 = 1. \quad (6.5)$$

Proof. Substitute (6.4) into (6.3). No further analytic input is used. $\square$

Remark 6.4 (Exactly what P1 gains). Theorem 4.1 proves uniqueness and gauge invariance of the rotational KMS extension for the abstract holomorphic net. Once a four-channel response resolution satisfying (6.1) is identified, equality of its weights is no longer an assumption: it is Theorem 4.1 plus Lemma 6.2. The factor $1/4$ is therefore a theorem-conditional consequence of index four and transitivity.

What is *not* proved is that the TFPT determinant-line response is represented by projections p_r of this kind, or that its physical coefficient is the channel response R_r . An outer fixed-point inclusion of factors does not supply four central local sector projections: the grading subspaces are not central summands. That operator identification is the joint P1.INDEX.KMS.01 and ALPHA.QUILLEN.EXACT.01 bridge. The angular normalization $\beta_{\text{angle}} = 2\pi$ is the geometric Bisognano–Wichmann/modular input, not a consequence of the group order. Accordingly, AX.P1.01 remains an axiom; (6.5) is a theorem-conditional identity, not an unconditional derivation of P1.

7 What this closes and does not close

SEAM.STATE.RPMIXING.01, continuum leg.

For an abstract holomorphic conformal net satisfying the hypotheses, the rotational KMS extension theorem is complete: there is one extension, its expectation is canonical, and both local relative-commutant mixing and thermal-sector mixing are excluded. The finite v995 computation is exactly consistent with this result. Applying it to the TFPT collar remains conditional on the continuum-net identification in the box below. The ledger’s continuum leg therefore stays open.

ALPHA.QUILLEN.EXACT.01.

The theorem supplies the rigidity premise: once the $U(1)$ determinant-line functional is identified with the invariant modular response of this inclusion, there is no second KMS state or nontrivial relative-commutant direction that can alter its sector weights. It does not prove the Bismut–Freed/Quillen identification, the zeta-regularized variation, or the compensation normalization. That contract remains open.

AX.P1.01.

The arithmetic identity $1/(4 \cdot 2\pi) = 1/(8\pi)$ and the equidistribution implication are proved under explicit operator and geometric hypotheses. The identification of that response with the primitive seam coefficient is not. Thus P1 stays typed as an axiom and gains a theorem-conditional reading, not a marker upgrade.

8 Remaining gap

Remaining gap (exact statement). For the TFPT $M = 1$ QWZ collar with its four quarter-twisted deck labels, prove that the scaling-limit local von Neumann algebras are the hyperfinite type-III₁ factors of a conformal net and identify, as conformal nets with vacuum, common stress tensor, rotation dynamics, and $\mathbb{Z}_4$ action, the resulting concrete inclusion with

$$\mathcal{A} = \mathcal{B}^{\mathbb{Z}_4} \subset \mathcal{B}, \quad [\mathcal{B} : \mathcal{A}] = 4, \quad \mathcal{A}(I)' \cap \mathcal{B}(I) = \mathbb{C}, \quad \mu_{\mathcal{B}} = 1, \quad c_{\mathcal{B}} = 8.$$

The identification must construct the non-bilinear spin/twist fields, verify relative locality and the commutative index-four Q-system, and show that the extension is the holomorphic $(E_8)_1$ net rather than merely matching its finite covariance, fusion, and index arithmetic. For the P1/Quillen application, the same identification must additionally show that the normalized determinant-line modular response is the additive transitive four-channel response of Lemma 6.2, with $\beta_{\text{angle}} = 2\pi$.

This is the MMST/Osborne–Stottmeister leg beyond fermion bilinears: the existing theorem controls the bilinear $\mathfrak{so}(16)_1$ currents, but not the spinor extension fields that generate $(E_8)_1$. Finite factoriality $Z(M_{256}) = \mathbb{C}$, the exact covariance (3.7), the $33 \rightarrow 0$ KMS rank test, and the $196608 \rightarrow 0$ invariant-extension test do not prove this type-III₁ scaling-limit identification.

Conclusion

The abstract operator-algebraic theorem is closed at its natural strength. Existence comes from the vacuum-sector rotational Gibbs state of the holomorphic extension. The index-four group average is the unique expectation by the finite-index Radon–Nikodym theorem and irreducibility. Takesaki’s theorem gives the explicit modular restriction and intertwining chain. Gauge-invariant extension uniqueness is an elementary averaging identity, while unconditional rotational KMS uniqueness requires both KLM ($\mu = 1$ leaves only the vacuum sector) and Longo–Tanimoto (rotational KMS extreme points are sector Gibbs states). Araki’s theorem identifies the exact frozen finite covariance and explains the finite KMS rank collapse.

The remaining issue is not another uniqueness argument. It is the construction and identification of the concrete TFPT collar scaling-limit net, including its non-bilinear extension fields and its determinant-line response. That issue is isolated in the single box above and lies outside Theorem 4.1.

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